Experiments and A/B Testing

Run a test that tells the truth, read it without fooling yourself, then decide.

WHY TEST AT ALL

⅓ • ⅓ • ⅓

Microsoft: a third of experiments help, a third do nothing, a third hurt

10-20%

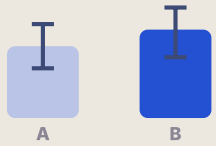
of Google and Bing experiments are positive

9 in 10

Booking.com: how often its ideas are wrong

THE WORKFLOW · SIX STEPS, IN ORDER

1 Randomise



⚠ a dashboard can't show cause

2 Write the brief

Hypothesis + reason · **OEC** · guardrails · **unit** · window · decision rule

⚠ success defined after the data

3 Size it

halve the effect → ~4× sample

⚠ guardrails and segments are under-powered

4 Read it in order

Health (split) → result + interval → guardrails → segments

⚠ peeking: 5% false alarms → 26%

5 Decide by the signed rule

ship · kill · iterate

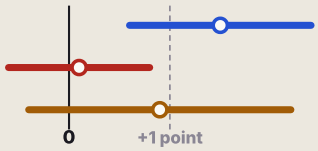
⚠ iterate with no date or budget

6 Or don't test

Low traffic, irreversible, unethical, or groups that interact: **staged rollout**

⚠ forcing an invalid test

READING THE INTERVAL · THE DECISION RULE, DRAWN



SHIP	the whole range is above zero (and guardrails hold)
STOP	the range rules out an effect worth having
ITERATE	anything in between

Each bar is a 95% interval for the effect. If it includes zero, you can't rule out **no effect at all**.

THREE RULES

Randomise the unit that would leak.

Check the split before the result.

An unplanned segment is the next hypothesis.

WHEN THESE MISLEAD YOU

A passed small test: volunteers, no control: not proof

A sample ratio mismatch: stop, find the leak, re-read

An AI-drafted readout: can skip the health check

A flat result: information, not failure

Decide what success means before the data arrives.

Sources: Kohavi & Thomke, HBR (2017); Kohavi, Tang & Xu (2020); Thomke on Booking.com, HBS Working Knowledge; Evan Miller, "How Not to Run an A/B Test".